Kaveri River Basin Water Quality Study 2025

Study scope. This study monitored 18 stations along the Kaveri River, from Talakaveri in Karnataka to Poompuhar in Tamil Nadu, with samples collected monthly from January to December 2025. The study was conducted by the Central Water Commission together with the State Pollution Control Boards. The parameters measured were pH, dissolved oxygen, biochemical oxygen demand, total coliform, turbidity, and nitrate.

pH results. The pH ranged from 6.8 to 8.4 across all stations, within the acceptable range of 6.5 to 8.5. The lowest pH of 6.8 was recorded at Station 14, Erode, in August. The highest pH of 8.4 was recorded at Station 3, Mysuru, in March.

Dissolved oxygen. Dissolved oxygen ranged from 4.1 to 8.9 milligrams per litre. 14 of the 18 stations met the standard of at least 5 milligrams per litre for bathing water. The worst dissolved oxygen level, 4.1 milligrams per litre, was recorded at Station 14, Erode, in September. The best level, 8.9 milligrams per litre, was recorded at Station 1, Talakaveri, in January.

Biochemical oxygen demand and coliform. Biochemical oxygen demand ranged from 1.2 to 9.8 milligrams per litre. The limit of 3 milligrams per litre was exceeded at 6 stations, all downstream of urban centres. Total coliform peaked at 24,000 MPN per 100 millilitres at Station 15, Tiruchirappalli, in October, against a limit of 500 MPN per 100 millilitres.

Nitrate and turbidity. Nitrate ranged from 2.1 to 18.6 milligrams per litre, below the drinking-water limit of 45 milligrams per litre at all stations. Turbidity peaked at 68 NTU during the July monsoon at Station 9, Srirangapatna. The dry-season average turbidity was 12 NTU.

Recommendations. The study makes four recommendations. First, upgrade sewage treatment at Erode and Tiruchirappalli. Second, create riparian buffer zones of 30 metres along urban stretches. Third, install real-time monitoring at 6 priority stations by June 2026. Fourth, launch a community reporting app. The estimated cost of the recommendations is 86 crore rupees over three years.